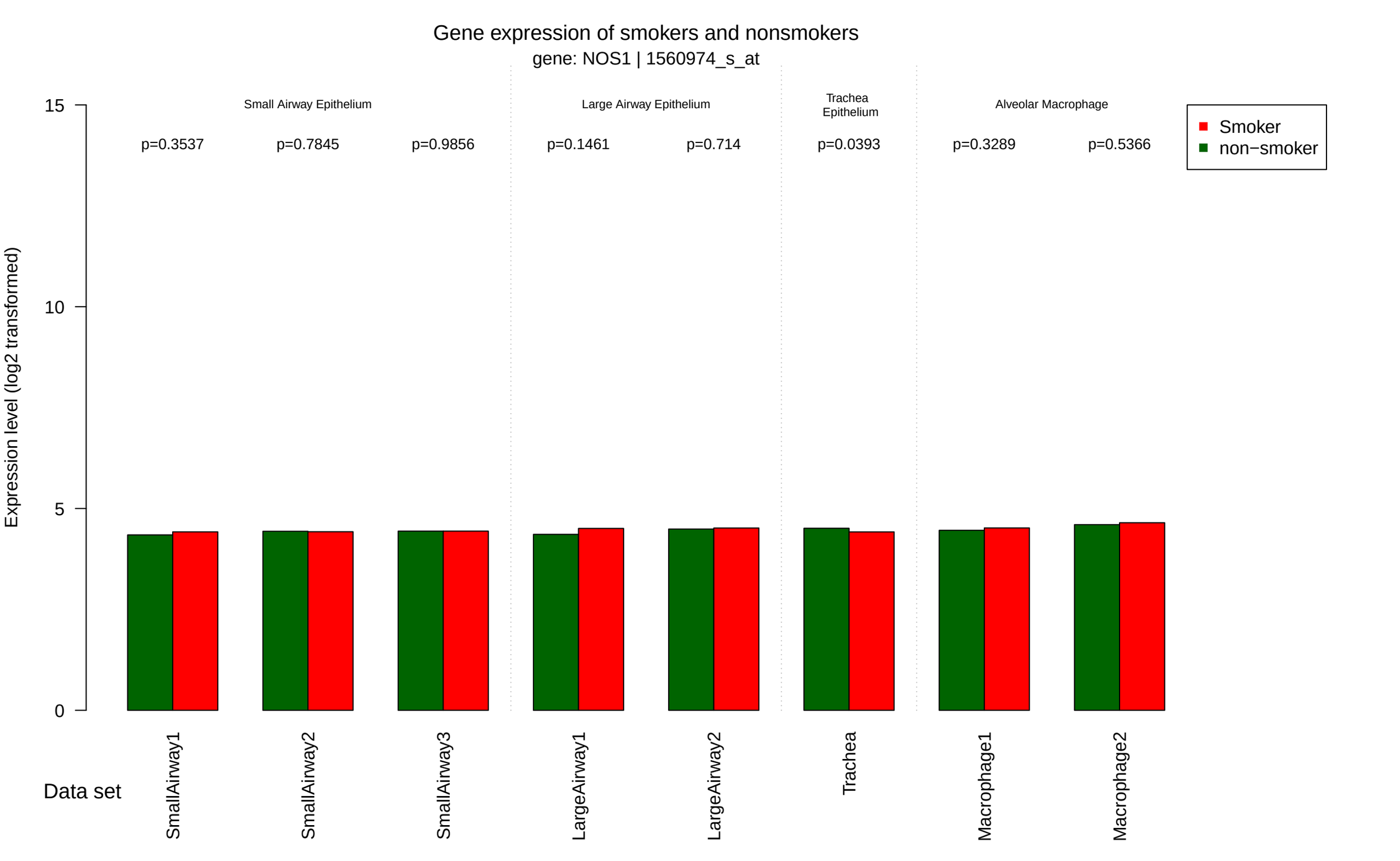
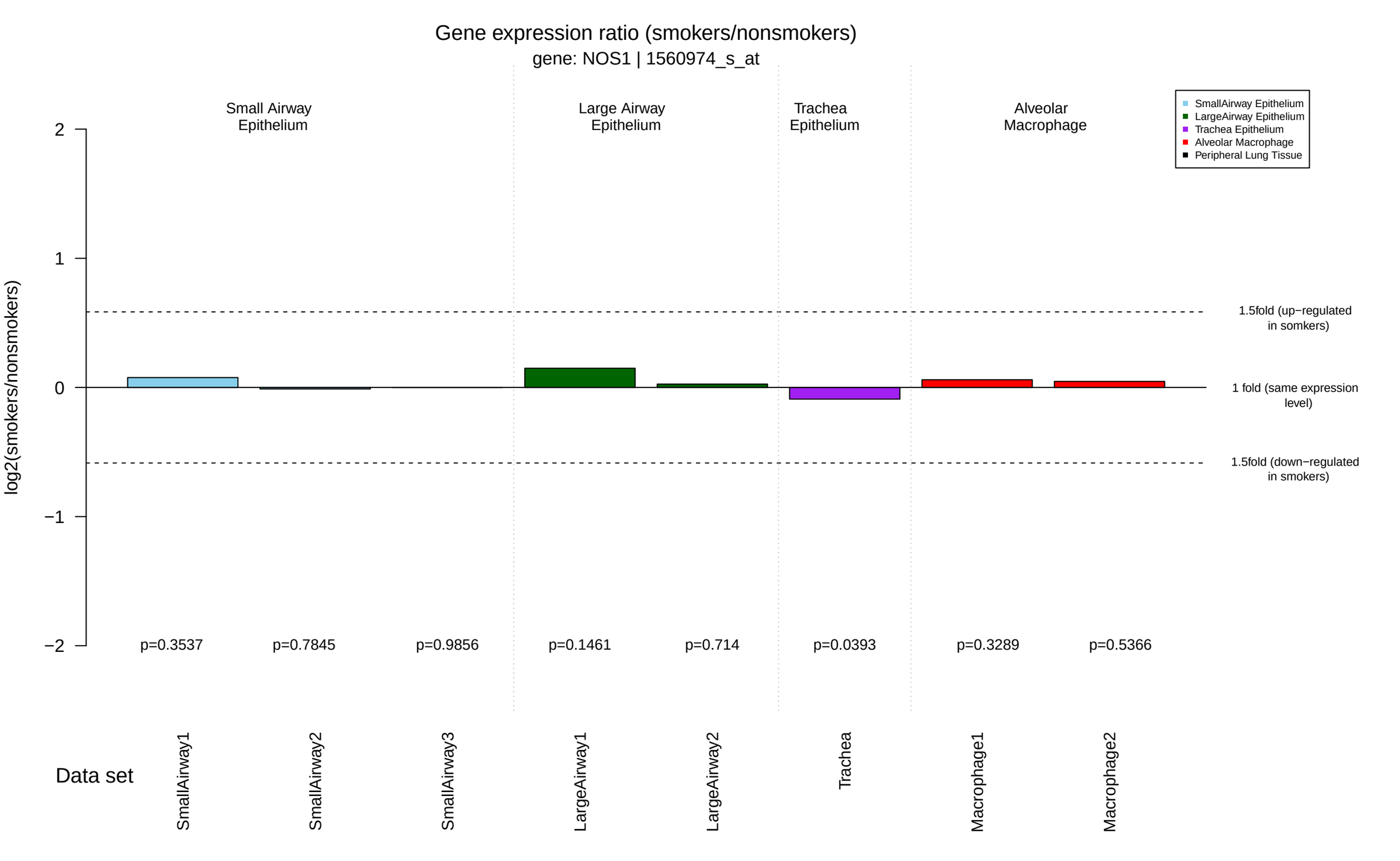
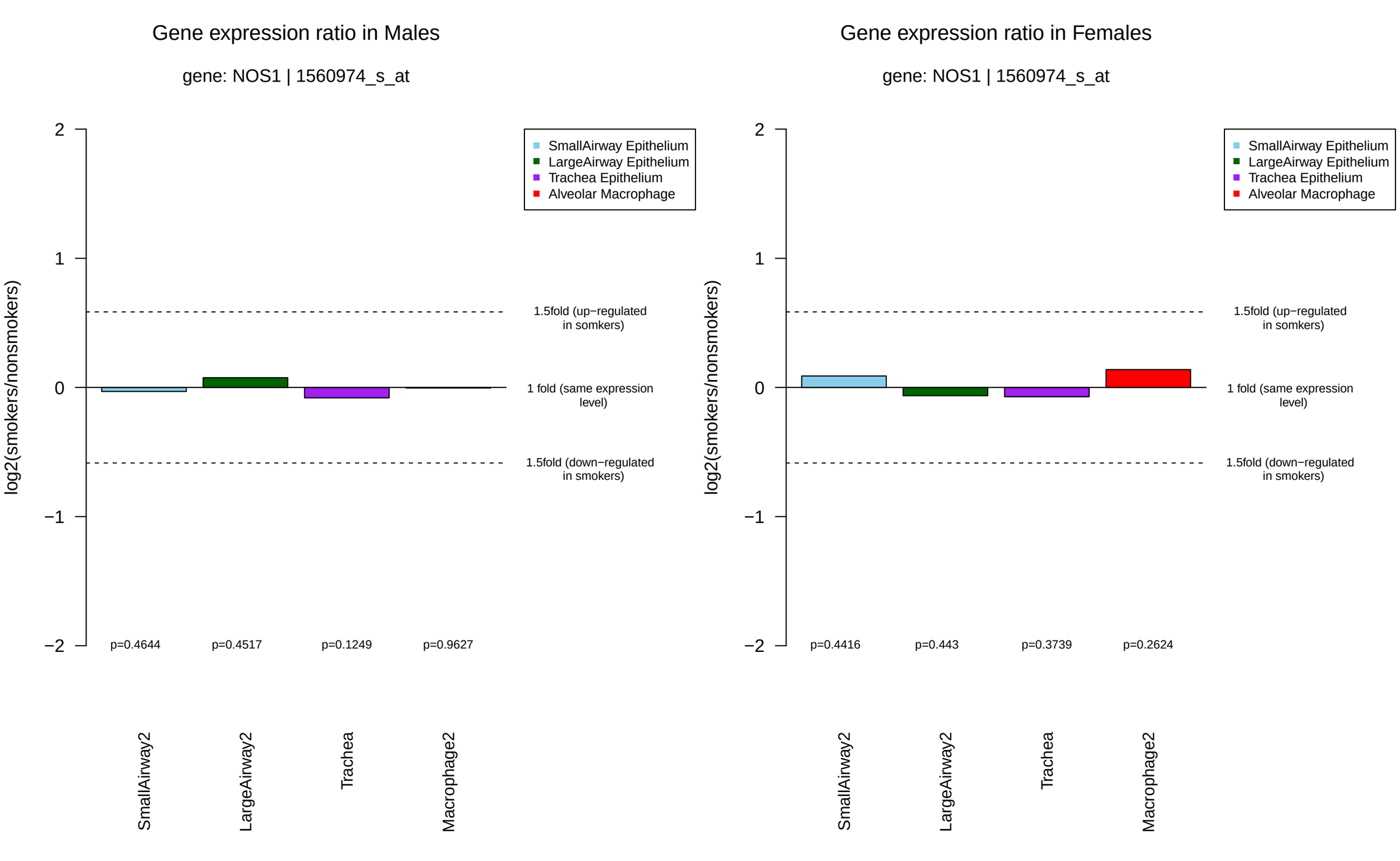
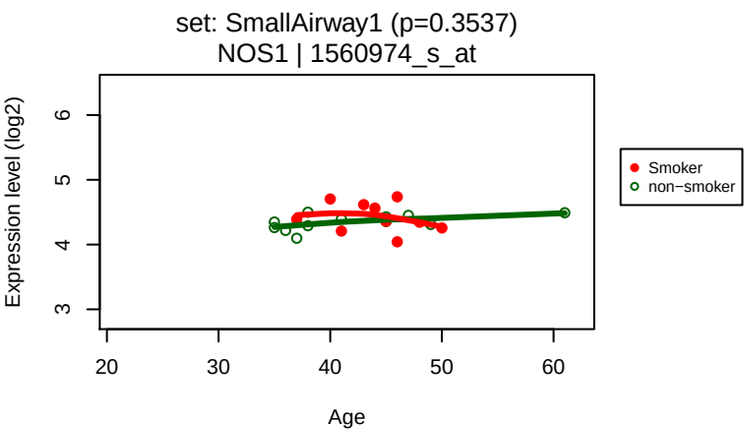
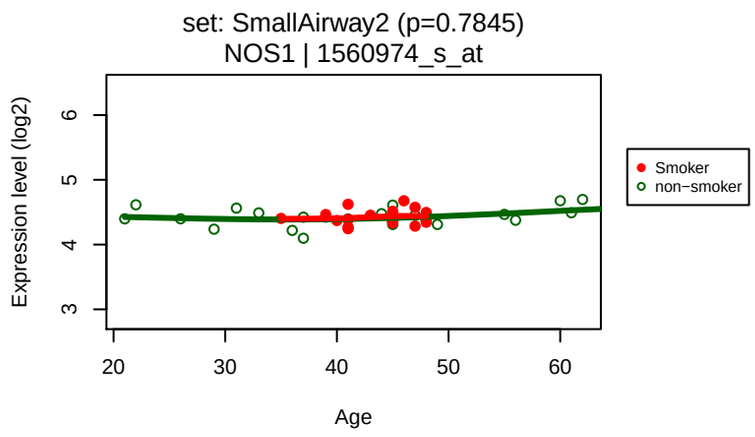
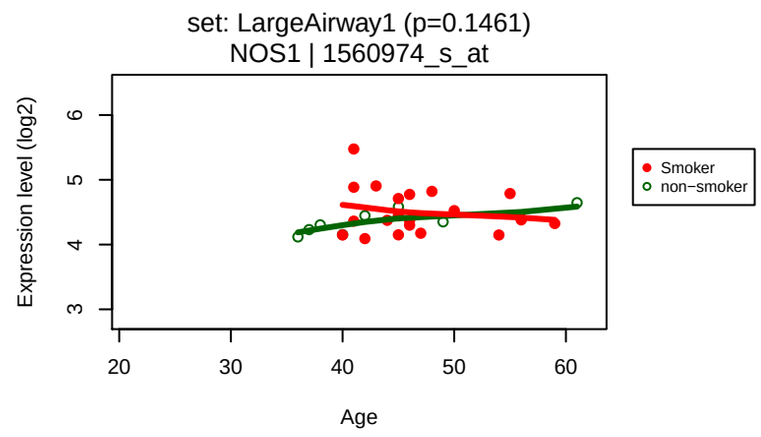

Gene expression of smokers and nonsmokers
gene: NOS1 | 1560974_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3537
p=0.7845
p=0.9856
p=0.1461
p=0.714
p=0.0393
p=0.3289
p=0.5366
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

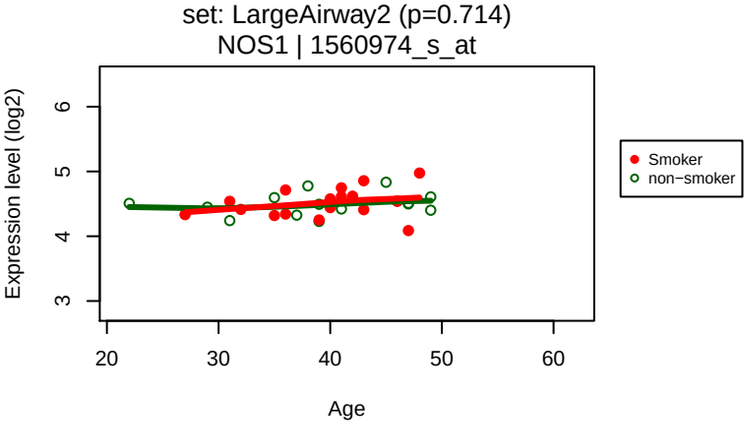

Gene expression ratio (smokers/nonsmokers)
gene: NOS1 | 1560974_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3537
p=0.7845
p=0.9856
p=0.1461
p=0.714
p=0.0393
p=0.3289
p=0.5366
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

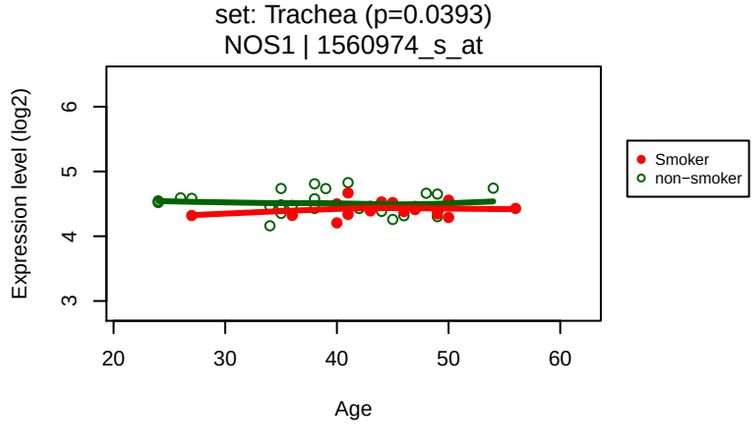

Gene expression ratio in Males
gene: NOS1 | 1560974_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4644
p=0.4517
p=0.1249
p=0.9627
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NOS1 | 1560974_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4416
p=0.443
p=0.3739
p=0.2624
SmallAirway2
LargeAirway2
Trachea
Macrophage2

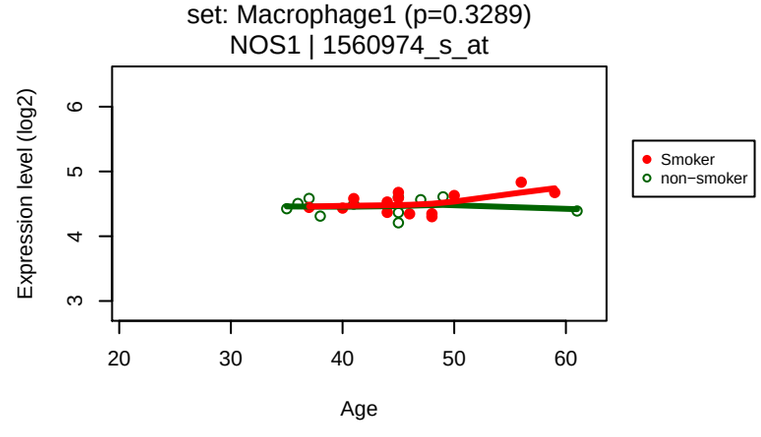

set: SmallAirway1 (p=0.3537)
NOS1 | 1560974_s_at
Expression level (log2)
Age
Smoker
non−smoker

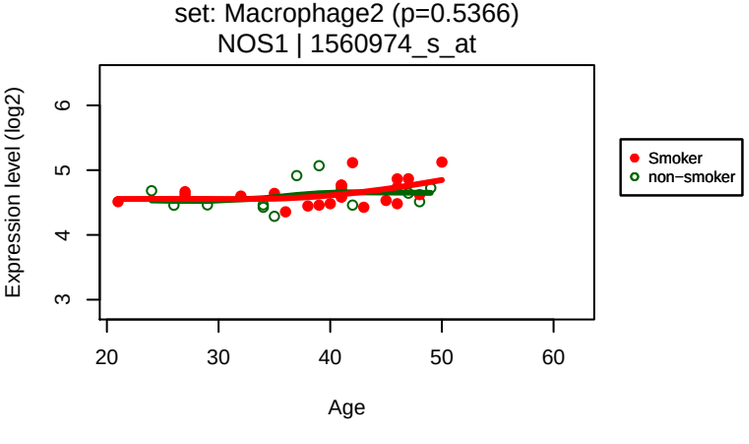

set: SmallAirway2 (p=0.7845)
NOS1 | 1560974_s_at
Expression level (log2)
Age
Smoker
non−smoker

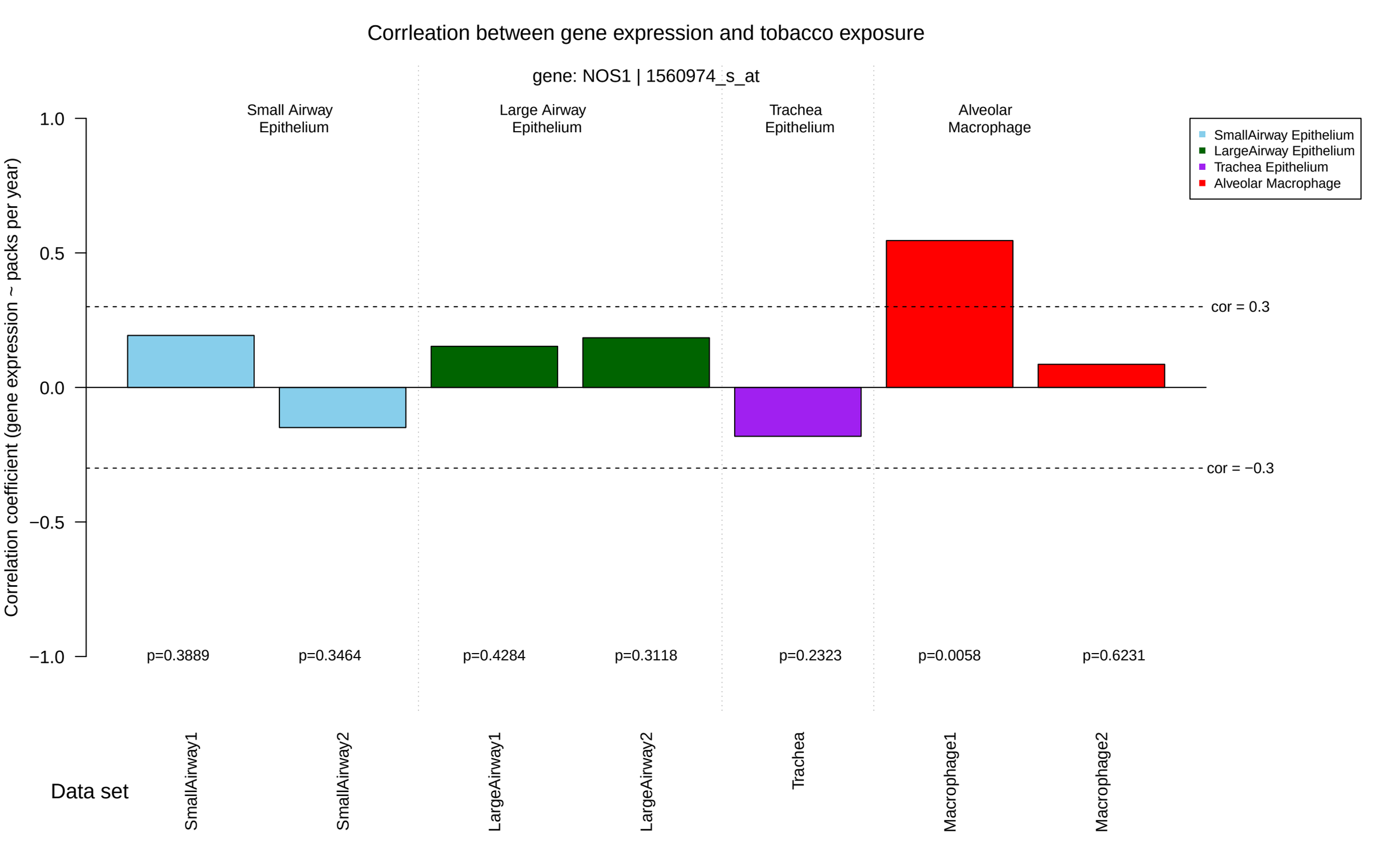

set: LargeAirway1 (p=0.1461)
NOS1 | 1560974_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.714)
NOS1 | 1560974_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0393)
NOS1 | 1560974_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3289)
NOS1 | 1560974_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5366)
NOS1 | 1560974_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NOS1 | 1560974_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3889
p=0.3464
p=0.4284
p=0.3118
p=0.2323
p=0.0058
p=0.6231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2